DSA03-Natural Language Processing

List of Lab Program

4. Implement a finite-state machine for morphological parsing. In this example, we will create a simple machine to generate plural forms of English nouns using python.

Code:

```
def plural_noun(word):
    # State 0: Start state
    state = 0
    # State 1: Noun ends with s, sh, ch, x, or z
    if word.endswith(("s", "sh", "ch", "x", "z")):
        state = 1
    # State 2: Noun ends with consonant + y
    elif word.endswith("y") and len(word) > 1 and word[-2].lower() not in "aeiou":
        state = 2
    # State 3: Other nouns
    else:
        state = 3
    # Generate plural form based on state
    if state == 1:
        return word + "es"
    elif state == 2:
        return word[:-1] + "ies"
```

```
    else:
        return word + "s"
word = input("Enter a noun: ")
print("Plural form:", plural_noun(word))
```

Output:



